

agent = model + harness; the harness is everything below

## THE MODEL: reasoning only (Claude session, Mode B) — no memory between calls, no ground truth

### 1 run the loop

burst-walkthrough protocol: RUN → PRESENT → GATE → LITERATURE → DISTILL

dispatcher: roster, gates, order at task intake

prototype chains: fit, merge, products, temporal, assembly

queue manager + typed workflows (A17)

### 2 execute tools

spectral engine: 24 models × every block, validity + rail columns

Bayesian blocks (3ML native) + significance merge

Stage-1 approval: detectors, background, source — stamped

figures, temporal chain, SED family, per-burst report

### 3 manage context

skill files (27) indexed by the step ledger; lessons with IDs

skill-reader: the binding checklist at every step open

standing contracts in the PI's quoted words (S-, R-items)

memory files + notes; fresh-context sub-agents (isolate)

### 4 persist state

state board derived from DISK evidence, never from memory

identity stamps, receipts, sidecars, commit pin

object schemas (10) + validation test; action trace with model id

first-class actions: intent → preflight → commit → receipt stamps

### 5 enforce boundaries

hooks: no-ship (sha), dispatch plan, raw-data guard

producer ≠ verifier ≠ approver; fresh-context verifiers

fit-table auditor, tie rule, count coordinates, edge stamps

loop caps, spend ledger, post-tool audit, pre-commit, sandbox

solid: DEPLOYED

dashed: PROPOSED

census 2026-09-02; each box maps to rows of Table T1

PI gate: every step

approval identity-stamped; never fabricated